



AGSHARAN SAKTHITHARAN

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Electrical & Electronic Engineering graduate with hands-on experience spanning software development, embedded, and IoT product development. Comfortable working across the full stack — from microcontroller firmware and communication protocols to backend APIs, databases, and end-to-end system integration and testing.

CORE COMPETENCIES

Systems Engineering: System architecture, requirements analysis, testing & validation, technical documentation, cross-functional teamwork
Embedded & IoT: ESP32/STM32 firmware, FreeRTOS, MQTT/Modbus RTU, PCB design, sensor & hardware integration
Software Engineering: Python, REST API design (FastAPI), SQLite/relational databases, Git, Docker, Linux, automation scripting

EDUCATION

Bachelor of Science in Electrical & Electronic Engineering — Second Class Upper Division Oct 2022 – Sep 2026
Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka
– Graduating with Second Class Upper Division Honours | Academic requirements completed, awaiting graduation.
– Final Year Project: Smart Home Energy Management System (SHEMS) — locally-hosted full-stack IoT platform
– Member, Rotaract Club of SLIIT (2023–2025)
Cambridge Advanced Level – Physical Sciences 2010 – 2022
OKI International School, Wattala, Sri Lanka
– Awarded Best Overall Results | Secretary, Science Association (2020–2022)

EXPERIENCE

IoT & Embedded Engineer Intern Jun 2025 – Sep 2025
SLTMobitel Digital Labs, Colombo, Sri Lanka
– Developed microcontroller firmware for live IoT product development involving MQTT, MQTTS, and Modbus RTU protocols
– Implemented IR signal learning systems with EEPROM storage and browser-based configuration via AsyncWebServer
– Contributed to PCB design review, BOM preparation, and PCB fabrication vendor evaluation
– Owned end-to-end system validation for four industrial monitoring devices — integration testing, calibration, and field deployment
Electrical Engineering Intern Jun 2024 – Sep 2024
Kelani Cables PLC, Kelaniya, Sri Lanka
– Assisted in preventive and breakdown maintenance of cable manufacturing machinery including motors, DC drives, & VFDs
– Performed insulation resistance testing (Megger), earth resistance testing, and QA high-voltage testing of armored cables
– Gained exposure to PLC wiring, relay ladder logic, and industrial panel wiring

PROJECTS

Smart Home Energy Management System (SHEMS) | *Raspberry Pi 5 · FastAPI · SQLite · Python · Docker · Cloudflare Tunnel · PWA*
– Designed and built the backend system architecture — REST API server (FastAPI), relational database schema (SQLite), and Python automation scripts
– Integrated an MQTT data pipeline with a PWA dashboard, deployed via Docker Compose with Cloudflare Tunnel for remote access
Industrial Machine Monitoring System | *ESP32 · MAX6675 · K-type Thermocouple · MQTT · PCB Design*
– Designed and deployed four field-ready industrial monitoring devices for tracking machine temperature and power
– Assisted design and review of PCB layouts, assembled devices, calibrated sensors, and performed end-to-end system validation
Smart AC Controller | *ESP32 · MQTT/MQTTS · IR Communication · AsyncWebServer*
– Implemented IR signal capture/decoding with EEPROM storage and a web-based configuration interface
Egg Crack Detection System | *Raspberry Pi 5 · Python · TensorFlow/Keras · VGG16 · OpenCV*
– Trained a VGG16 CNN for crack classification with IR-triggered conveyor control; 72.3% accuracy on prototype dataset
Semantic-Aware Image Compression (DSP Assignment) | *MATLAB · DCT · Vector Quantization · LBG Algorithm · OCR*
– Implemented OCR-based semantic masking for text-preserving block DCT compression; near-lossless text reconstruction
Other Projects: Line Follower Robot (ESP32, Fuzzy Logic); Server Room Environmental Monitoring System (ESP32, LabVIEW).

TECHNICAL SKILLS

Programming & Software:	Python, Embedded C/C++, FastAPI, SQLite, REST APIs, Git, Docker, Linux, MATLAB
Systems & Protocols:	MQTT/MQTTS, Modbus RTU, UART, SPI, I2C, WebSocket, IR Communication
Microcontrollers & Firmware:	ESP32, STM32, FreeRTOS, Arduino IDE, PlatformIO
Hardware & Electronics:	PCB Design (EasyEDA), Circuit Assembly & Soldering, Sensor Interfacing, AutoCAD
Engineering Tools:	LabVIEW, Quartus II (FPGA/VHDL), NI Multisim, Proteus, Raspberry Pi, Cloudflare Tunnel
Languages:	English (Fluent), Tamil (Native), Sinhala (Conversational)

REFERENCES

Available upon request.